

# ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Models

Email: [atu1@umd.edu](mailto:atu1@umd.edu)

LinkedIn: [linkedin.com/in/allentu](https://www.linkedin.com/in/allentu)

Website: [tuallen.github.io](https://tuallen.github.io)

Google Scholar: [Allen Tu](#)

## SELECTED PUBLICATIONS

---

1. **Allen Tu\***, H. Ying\*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *CVPR*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *CVPR*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *FG*, 2026 (**Oral Presentation**).
4. A. Hanson\*, **Allen Tu\***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’. *CVPR*, 2025.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’. *CVPR*, 2025.

\* denotes equal contribution.

## EDUCATION

---

**University of Maryland, College Park**

*Ph.D. Student, Computer Science*

College Park, MD

January 2025 – May 2028

- Advised by Professor Tom Goldstein; collaborating with Professor Matthias Zwicker

*B.S./M.S. in Computer Science, Minor in Statistics*

August 2019 – December 2024

## RESEARCH EXPERIENCE

---

**Amazon**

*Applied Science Intern*

Bellevue, WA

June 2026 – October 2026

- Researching 3D digital twins for physical AI in the Last Mile Delivery and Automation org

**University of Maryland Institute of Advanced Computer Studies**

*Graduate Research Assistant*

College Park, MD

August 2023 – Present

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude ([WRIVA](#)) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over  $10\times$  fewer primitives while preserving visual fidelity [1, 4, 5]
- Incorporated uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [2]

**Systems & Technology Research**

*Computer Vision Research Intern*

Arlington, VA

June 2022 – January 2026

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range ([BRIAR](#)) program
- Introduced an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at  $1e-3$  FMR using an encoder trained solely on public data [3]

## SELECTED HONORS & SERVICE

---

[FG 2026 Doctoral Consortium](#)

**Workshop Organizer:** [SPAR-3D: Security, Privacy, & Adversarial Robustness in 3D Generative Vision Models](#), CVPR 2026